

Engineering Scope Document – Cooperative Management & Governance System | Embu County Government

Engineering Scope Document

Cooperative Management & Governance System

Embu County Government · Phases, Dependencies & Risk Register

	Project Type	**Fixed-price contract**		**Timeline**	**14 weeks**	
Total Budget	**KES 1,700,000**		**Client**	**Embu County Gov.**		
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1. Executive Summary

This document defines the full engineering scope for the Cooperative Management and Governance System commissioned by Embu County Government. The system will replace manual cooperative oversight processes with a secure, enterprise-grade web platform covering five core modules: Role-Based Access Control, Field Operations, Cooperative Registry, Document Management, and Governance/Election Tracking.

The delivery is structured over five sequential phases across 12-14 weeks, with payment milestones tied to concrete deliverables. This scope document captures each phase in detail, all inter-phase and external dependencies, a risk register with mitigations, and a list of open unknowns that must be resolved during Phase 1 to prevent downstream blockers.

2. Payment Milestones

	Milestone		**Trigger**		**Deliverable Gate**		**Amount (KES)**	
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	M1 – 30%		Contract execution		Prior to UI/UX design commencement		**510,000**	
	M2 – 30%		Phase 2 complete		Staff portal, cooperative directory, and document upload pipeline live on staging		**510,000**	
	M3 – 20%		Phase 3 complete		Governance engine, election logs, and compliance checker deployed to staging		**340,000**	
	M4 – 20%		Final handover		UAT sign-off, staff training complete, production deployment, source code transfer		**340,000**	

3. Project Phases

The delivery lifecycle is broken into five sequential phases. Each phase has a defined entry condition (what must be true before it starts), key deliverables, and an exit condition (what must be true before the next phase starts).

	Phase		**Title**		**Description**		**Key Deliverables**	
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	P1		Wk 1-3		**Design, Schema ****&**** RBAC**		Produce all technical specification documents, interactive UI/UX wireframes, PostgreSQL schema, and stand up the county staff access-management module (Module 1) plus the field operations portal (Module 2).	
							• Technical spec doc • UI/UX wireframes • Database schema • Module 1: RBAC • Module 2: Field ops & leave	
	P2		Wk 4-7		**Core Development ****&**** DMS**		Build the full cooperative registry, member database, and the multi-tier document approval pipeline. Modules 3 and 4 are tightly coupled – both must ship together for the Milestone 2 payment to trigger.	
							• Module 3: Cooperative registry • Module 4: Document mgmt system • 3-tier approval state engine • Cloud file storage config	
	P3		Wk 8-10		**Governance Engine**		Implement election lifecycle tracking, the 1/3 gender rotation compliance algorithm, candidate application interface, committee structural mapping, and AGM archival tools. Deploy to staging for client review.	
							• Module 5: Governance • Election lifecycle engine	

- 1/3 compliance checker • Staging deployment |
- | ****P4**** Wk 11-12 | ****QA, Security ****&**** Optimisation**** | End-to-end security testing across all modules, data sanitisation, role-permission boundary checks, pre-signed URL expiry testing, and performance optimisation for large member data tables. | • Security audit report • Bug-fix and patch log • Performance benchmarks |
- | ****P5**** Wk 13-14 | ****Migration, Training ****&**** Handover**** | Run bulk Excel/CSV member import scripts, conduct on-site training workshops for county staff, complete UAT sign-off, and push to the live production environment. Full IP transfer at final payment. | • Imported member data • On-site training workshop • Admin manual & video guides • UAT sign-off • Source code transfer |

****Phase entry and exit conditions****

- Phase 2 may not begin until: Phase 1 governance logic spec is signed off by Embu County Director.
- Phase 3 may not begin until: Module 3 (cooperative registry) and Module 4 (DMS) are both live on staging.
- Phase 4 may not begin until: All five modules are integrated and accessible in staging.
- Phase 5 may not begin until: QA sign-off is complete and no critical or high-severity bugs remain open.

****4. Dependencies****

****4.1 County Dependencies (External)****

The following items are the responsibility of Embu County Government. Delays in any of these will directly extend the project timeline. These should be codified as contractual obligations.

Dependency	**Owner**	**Required By**	**Consequence if Late**
Named IT liaison confirmed in writing	County Director	**Before Week 1**	Infrastructure and domain decisions blocked
Clean member data in CSV/Excel templates provided	County IT/Cooperatives Dept	**By Week 11**	Phase 5 import scripts cannot be built or tested
Validation/UAT feedback turnaround within 5 business days	County IT Liaison	**Ongoing (Phases 3-5)**	Testing loops compress; go-live date slips
Staff availability confirmed for training workshops	County HR / Director	**By Week 12**	Phase 5 cannot complete; Milestone 4 payment blocked
Production server / cloud account access provided	County IT Liaison	**By Week 13**	Production deployment cannot proceed
Domain name and DNS control confirmed	County IT Liaison	**By Week 13**	Go-live URL cannot be configured

****4.2 Technical Dependencies (Internal)****

- RBAC (Module 1) must be stable and deployed to staging before any other module can ship – all role-permission checks depend on it.
- Module 3 (cooperative registry) must be live before Module 4 (DMS) can link documents to specific cooperatives.
- The Phase 1 PostgreSQL schema must be signed off by the client before Phase 2 development begins – schema changes mid-build are the single most expensive rework scenario.
- Cloud object storage must be configured and access credentials provided before DMS pipeline testing can begin in Phase 2.

- A staging environment must be provisioned by Week 8 at the latest to support Phase 3 deployment and client review.

- Governance logic specification (1/3 rotation rules, election term lengths, committee structures) must be frozen in a written spec by end of Phase 1.

5. Risk Register

Risks are rated High, Medium, or Low based on likelihood combined with impact to timeline and cost. All mitigations are actionable and assigned to a phase.

Risk	**Level**	**Mitigation**
County delivers dirty or incomplete member data late	**High**	Provide CSV templates in Week 1. Set a hard data-delivery deadline of Week 11 as a contractual obligation tied to Milestone 4. Developer cannot be held liable for delay caused by late data.
Scope creep on governance engine (1/3-rule edge cases, custom committee structures)	**High**	Freeze governance logic specification by end of Phase 1. Any new rules post-freeze go into a formal change-request log with cost and time estimates before work begins.
Multi-tier document approval state machine complexity underestimated	**Medium**	Prototype the state engine in Week 3 as part of Phase 1 technical spec output. Add a 3-day development buffer in Phase 2 schedule.
Pre-signed URL expiry logic creates security gaps or UX friction on slow connections	**Medium**	Define URL TTL policy (recommended: 15-minute expiry) in Phase 1 spec. Test download flows under throttled connectivity during Phase 4 security audit.
IT liaison unavailable or slow to respond, blocking infrastructure decisions	**Medium**	Named liaison must be confirmed in writing before contract execution. Escalation path to County Director if liaison is unresponsive for more than 48 business hours.
UAT feedback arrives late, compressing Phase 5 or missing go-live date	**Medium**	Implement rolling UAT during Phase 4 rather than a big-bang review at the end. County Director and sub-county leads test each module as it passes QA sign-off.
Cloud infrastructure costs exceed county budget post-launch	**Low**	Provide a cost projection (storage per PDF × 200 cooperatives × avg. document count) in the Phase 1 technical spec so the county can provision budget before go-live.
Staff turnover during 90-day warranty period creates knowledge gaps	**Low**	Deliver a written admin manual and recorded video walkthroughs as part of the Phase 5 handover package, in addition to the live training workshops.

6. Open Unknowns – Resolve in Phase 1

The following items are currently undefined and will produce blockers or costly rework if not resolved before Phase 2 development begins. These must be addressed during the Phase 1 discovery and specification process.

- **Geographic structure:** Exact number of sub-counties and wards to be loaded as dropdown options – directly affects database schema design and cannot be changed without migration after Phase 2 starts.

- **Cooperative manager access model:** Whether cooperative managers receive a direct login to the system or whether county field staff enter data on their behalf. This changes the Module 1 role count and the entire permission matrix.

- **PDF storage policy:** Maximum permitted file size per document upload, and the retention policy – active-only, or long-term archival with version history.

- **Offline/mobile entry requirement:** Whether the candidate election application interface needs to function offline or support low-bandwidth mobile entry by field staff during cooperative visits.

- ****Notification and alerting requirements:** ****Whether the system should send email or SMS alerts when committee terms are approaching expiry, when documents are pending approval, or when compliance violations are detected.**

- ****Production environment ownership:** ****Whether Embu County will provide its own cloud account (developer deploys into it) or whether the developer manages hosting with pass-through billing – this affects both the deployment architecture and ongoing cost accountability.**

****7. Governance Logic Specification (Module 5)****

This section documents the algorithmic rules that must be implemented in the governance engine. These rules must be confirmed and signed off by the County Director before Phase 3 development begins. Any rule not listed here that is requested after Phase 2 ends will be treated as a change request.

****7.1 Committee Term Tracking****

- Each management committee member has an `election_date` and a calculated `reelection_due_date`.

- Default term length: 3 years (to be confirmed by county). System must allow the Director to configure term length per value chain if required.

- The system flags a committee as 'Term Expiring' 90 days before any member's `reelection_due_date`.

- The system flags a committee as 'Term Expired – Action Required' on the day a member's term passes without a recorded re-election.

****7.2 One-Third Gender Rotation Rule****

- At any point in time, no single gender may hold more than two-thirds (2/3) of the elected management committee positions.

- The compliance checker runs automatically when a new committee composition is saved. It evaluates: $\text{total elected seats} \div \text{gender count per gender}$.

- Non-compliant committee compositions are flagged with a 'Compliance Violation' badge visible to the sub-county officer and Director. Submission is blocked until corrected or manually overridden by the Director with a logged justification.

- The rule applies to: Chairperson, Vice Chairperson, Secretary, Treasurer, and ordinary board members. The non-elected Executive Manager is excluded from the calculation.

****7.3 Election Lifecycle States****

- Upcoming: 90+ days before `reelection_due_date` – informational status only.

- Election Due: within 90 days – system generates notification (if alerting is enabled).

- Election Overdue: past `reelection_due_date` with no new election recorded – system flags for Director review.

- Candidate Application Open: manually opened by county staff; candidates may apply through the system.

- Election Recorded: new committee composition saved and compliance-checked.

****7.4 Data Migration Checklist****

The following data must be prepared by Embu County in the provided CSV templates before Week 11:

- Cooperative name, registration number, value chain, sub-county, ward – for all 200 cooperatives.
- Member legal name, national ID number, phone number(s), gender, share capital amount – per cooperative.
- Current committee composition: names, roles, election dates – per cooperative.
- Any existing documents (By-Laws, audit reports, AGM minutes) to be uploaded – as PDF files, labelled with cooperative name and document type.
- Field staff roster: full name, job group, designation, phone, email, sub-county assignment.

This document is a living scope reference. Changes to agreed scope must be raised as formal change requests with written sign-off from both parties.

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